Meeting 5D — Storage Layer and Diagnostics Layer Software Architecture

Digital Harmonium Workstation

Final Design Specification

Purpose of These Layers

The musical layers of the workstation are responsible for creating sound.

These layers are responsible for making the workstation reliable, maintainable, recoverable, and easy to develop.

The **Storage Layer** manages every piece of persistent information.

The **Diagnostics Layer** continuously monitors the health of the entire workstation.

Neither layer directly produces audio.

Neither layer performs synthesis.

Instead, they provide the infrastructure that allows every other subsystem to operate safely and predictably.

Layer 7 — Storage Layer

Philosophy

The Storage Layer answers one fundamental question:

> **"Where does information live, and how is it safely managed?"**

This layer isolates every file, memory allocation, configuration, and recording operation from the rest of the firmware.

No other subsystem should directly manipulate files or persistent storage.

Module 1 — VHL Library Manager

Purpose

Manages the collection of VHL instrument files stored on the SD card.

Responsibilities

- Discover available VHL instruments.
- Read instrument metadata.
- Verify VHL headers before loading.
- Maintain instrument library index.
- Provide instrument list to the user interface.
- Request loading of selected instruments.

Inputs

- SD Card
- User instrument selection

Outputs

- Selected VHL file
- Instrument metadata

Notes

This module never loads instruments into playback memory itself.

It only manages the instrument library.

Module 2 — Recording Storage Manager

Purpose

Stores and retrieves recorded performances.

Responsibilities

- Save new recordings.
- Load recordings.
- Delete recordings.
- Rename recordings.
- Browse available recordings.
- Organize recording files.

Stored Data

- Note events
- Timing information
- Performance metadata

Notes

This module stores musical performances rather than raw audio.

Module 3 — Configuration Manager

Purpose

Maintains all persistent workstation settings.

Responsibilities

- Save user preferences.
- Load configuration during startup.
- Restore default settings.
- Update configuration when settings change.

Typical Settings

- Master volume
- LCD contrast
- Startup instrument
- Default transpose
- Last selected taal
- Last selected tempo

Notes

Settings survive power cycles.

Module 4 — File System Manager

Purpose

Provides controlled access to the SD card file system.

Responsibilities

- Mount SD card.
- Detect insertion and removal.
- Browse directories.
- Verify file existence.
- Open files.
- Close files.
- Handle file errors.

Notes

No other module communicates directly with FATFS or SD storage.

This creates a single point of storage management.

Module 5 — Memory Manager

Purpose

Controls runtime memory allocation.

Responsibilities

- Allocate RAM.
- Release RAM.
- Track available memory.
- Prevent memory fragmentation.

- Report memory usage.
- Support future PSRAM expansion.

Managed Memory

- Instrument memory
- Recording buffers
- Temporary working buffers
- DMA buffers

Notes

Memory ownership is centralized here to prevent accidental corruption.

Module 6 — Cache Manager

Purpose

Improves performance by retaining recently used resources.

Responsibilities

- Cache recently loaded instruments.
- Reuse previously loaded data.
- Discard cached resources when memory becomes limited.
- Track cache usage.

Future Expansion

- Multiple instrument cache.
- Recently used recording cache.

Notes

Entirely optional during early development.

Module 7 — Backup / Export Manager

Purpose

Supports long-term preservation of user data.

Responsibilities

- Export recordings.
- Backup configuration.
- Restore backups.
- Copy user data.
- Support firmware migration.

Future Expansion

- USB export.
- Computer synchronization.

Notes

This module is not required for the first prototype but is reserved in the architecture.

Storage Layer Summary

The Storage Layer owns:

- Instruments
- Recordings
- Configuration
- Files
- Memory
- Cache
- Backups

Nothing else in the system directly manages persistent storage.

Layer 8 — Diagnostics Layer

Philosophy

The Diagnostics Layer continuously observes the workstation and evaluates whether every subsystem is functioning correctly.

Unlike other layers, diagnostics never generate sound or control features.

They simply monitor, measure, and report.

Module 8 — System Diagnostics

Purpose

Monitors the overall operating condition of the firmware.

Responsibilities

- Perform boot diagnostics.
- Monitor system health.
- Track processor activity.
- Verify successful initialization.
- Detect abnormal operating conditions.

Notes

Provides a global view of system status.

Module 9 — Hardware Diagnostics

Purpose

Verifies correct operation of all physical hardware.

Responsibilities

- Test keyboard matrix.
- Test LEDs.
- Test buttons.
- Test LCD.
- Verify SD card presence.
- Verify speaker output.
- Verify headphone detection.

Startup Tests

- LED sequence
- Audio beep
- Display initialization
- Input verification

Notes

Executed automatically during startup.

Module 10 — Audio Diagnostics

Purpose

Observes the real-time audio engine.

Responsibilities

- Detect audio underruns.
- Monitor DMA activity.
- Detect clipping.
- Measure active polyphony.
- Verify mixer stability.
- Observe playback performance.

Notes

Critical for low-latency debugging.

Module 11 — Performance Monitor

Purpose

Measures runtime performance.

Responsibilities

- CPU usage.
- RAM usage.
- Mixer execution time.
- ISR execution time.
- Audio processing latency.
- System throughput.

Notes

Used primarily during firmware optimization.

Module 12 — Error Manager

Purpose

Acts as the centralized repository for all system errors.

Responsibilities

- Receive error reports.
- Classify errors.
- Store current error state.
- Notify Display Manager.
- Support future recovery actions.

Typical Errors

- SD card missing
- Instrument load failure
- Memory exhaustion
- Invalid VHL file

- Audio failure

Notes

No module displays its own errors directly.

All errors pass through the Error Manager.

Module 13 — Logging Manager

Purpose

Maintains runtime event logs during development.

Responsibilities

- Log boot events.
- Log instrument loading.
- Log recording activity.
- Log playback activity.
- Log warnings.
- Log internal errors.

Output

- USB Serial
- Future debug interfaces

Notes

This module may be disabled in production firmware to reduce overhead.

Diagnostics Layer Summary

The Diagnostics Layer continuously observes:

- System health
- Hardware health

- Audio health
- Runtime performance
- Errors
- Development logs

It never changes workstation behavior directly.

It simply reports what is happening.

Final Storage Layer Modules

1. VHL Library Manager
2. Recording Storage Manager
3. Configuration Manager
4. File System Manager
5. Memory Manager
6. Cache Manager
7. Backup / Export Manager

Final Diagnostics Layer Modules

1. System Diagnostics
2. Hardware Diagnostics
3. Audio Diagnostics
4. Performance Monitor
5. Error Manager
6. Logging Manager

Meeting Summary

With this document, every major non-musical subsystem of the workstation has been identified and assigned a dedicated software module.

The Storage Layer ensures that instruments, recordings, configuration, and memory remain organized and reliable.

The Diagnostics Layer ensures that the workstation remains observable, debuggable, and maintainable throughout development and future expansion.

Together, these thirteen modules complete the infrastructure surrounding the musical core of the Digital Harmonium Workstation.

End of Meeting 5D — Storage Layer and Diagnostics Layer Software Architecture